

LASER THIEF DETECTION SYSTEM

Minor Project Report - Department of ECE

By: Akash Rathore & Kuldeep Patel

Guided by: Mr. Manish Purohit

Introduction & Project Objective

- • Optical perimeter intrusion security system.
- • Real-time beam interruption sensing using Laser & LDR.
- • Microcontroller threshold logic to trigger immediate auditory alert.
- • Cost-effective, high-reliability security prototype for residential & commercial protection.

Hardware Requirements

- • Laser Diode Module (650nm Optical Transmitter)
- • Light Dependent Resistor (LDR Receiver Module)
- • Controller Board (Embedded threshold detection)
- • Active Buzzer & Warning LED
- • Breadboard, Resistors, Jumper Wires & Power Supply

Working Principle & Detection Logic

- • Continuous optical beam establishes line of sight between Laser & LDR.
- • When path is unobstructed: LDR resistance is low, voltage exceeds threshold.
- • When intruder crosses beam: Light drops, LDR resistance surges.
- • Controller detects threshold trip within milliseconds and fires buzzer alert.

Applications & Future Scope

- • Perimeter fencing for homes, vaults, and laboratories.
- • Industrial warehouse boundary surveillance.
- • Future upgrade: IoT integration for GSM/SMS instant alert notification.
- • Multiple-beam triangulation for directional movement tracking.